HAISEM NAEEM

APPLIED AI ENGINEER

Stockholm, Sweden | +46 76 961 9249 | haisem.work@gmail.com
linkedin.com/in/haisem-naeem | github.com/haisem-codes | haisem-codes.github.io

PROFESSIONAL SUMMARY

Applied AI Engineer who moved from Pakistan to Sweden to pursue an MSc in Artificial Intelligence and Language at Stockholm University. Experienced in building agentic, retrieval and multimodal AI applications, including leading AI integration across three SaaS platforms at Metaviz. Stays current with emerging AI tools and AI-assisted development practices to deliver high-quality software more efficiently. Chose this degree to deepen the scientific foundations behind that work and is seeking Applied AI engineering opportunities alongside master's studies.

CORE COMPETENCIES

Agentic AI systems | LLM integration | Structured tool calling | Prompt engineering | RAG and hybrid retrieval | Guardrails | Human-in-the-loop workflows | Scenario evaluations | Prompt caching

PROFESSIONAL EXPERIENCE

Metaviz Lahore, Pakistan
AI Engineer | Full-time Jul 2025 - Aug 2026

- Led the complete agentic AI integration and backend development for Wanyumba (property), AfricaTravelHub (travel) and Handman/Getapro (trades), engineering PydanticAI assistants with LLM APIs, contextual system prompts and typed tools for domain workflows.
- Integrated pgvector and PostgreSQL full-text retrieval with reciprocal rank fusion in AfricaTravelHub and Handman, grounding conversational discovery in ranked inventory and professional matches. Configured model providers and prompt-cache keys in Wanyumba and Handman.
- Implemented topic guardrails in AfricaTravelHub and RBAC, capability gates and human-approved actions in Handman. Added Handman scenario evaluations for tool selection and workflow state, supported by token attribution, timing logs and tool audits.
- Engineered building-plan compliance across five code disciplines using LangGraph orchestration, OCR, hybrid retrieval and citation validation, supported by 132 FastAPI endpoints. Reduced manual review time by approximately 80% and average inference spend per document by approximately 40%.
- Delivered Coach multimodal habit coaching with OpenAI Realtime, LiveKit and photo verification, integrating conversational context with 21 Firestore collections. Reduced session token cost by approximately 50% through prompt caching.
- Developed Postura on-device ergonomic assessment using YOLOv8n/TensorFlow Lite, pose estimation and ROSA scoring; ported Python scoring to Dart and created comparisons against a 71-photo reference set.
- Built ICore Career Hub AI APIs for structured PDF/DOCX CV extraction, skill filtering and embedding-based candidate matching, with validation, retries and vector-profile synchronization.
- Implemented the DietApp AI recommendation API with output validation, retries and fallbacks; later replaced LLM inference with deterministic recommendation rules.
- Delivered voice-agent qualification, appointment-booking and CRM-update workflows for more than five business owners using Retell, GoHighLevel, n8n, APIs and webhooks.
- Contributed to Metaviz AI productivity tools for repository explanations, interview support, multilingual blogs and product copy, plus a real-time voice prototype.

WebBuggs Lahore, Pakistan
Python Developer Intern Jul 2024 - Jan 2025

- Built a FastAPI retrieval-augmented generation backend for real-time answers grounded in a private document corpus.
- Developed a Django backend for live multilingual transcription, subtitles and an AI meeting assistant.

EDUCATION

Stockholm University — Stockholm, Sweden
MSc Artificial Intelligence and Language — Aug 2026 - Jun 2028 expected

FAST-NUCES Lahore — Lahore, Pakistan
BS Computer Science — Sep 2021 - Jun 2025
CGPA: 3.19/4.0 | Dean's List in three semesters

PERSONAL PROJECTS

Aegis | AI Assisted Code Security Analysis
- Built a typed Python prototype combining Joern code graphs, static data-flow candidates and LLM-assisted reasoning. Added adversarial evidence checks, model-call budgets, provider-failure handling and SARIF reporting.

Notetaker | Lecture Transcription and Study Notes
- Built a React/FastAPI application using local MLX Whisper transcription and Claude/OpenAI note generation. Added Pydantic validation, targeted repair, partial-result recovery, background jobs and PDF export.

Virtual Tour | AI Itinerary Video Generator
- Developed a seven-stage BullMQ pipeline combining structured LLM extraction, scene planning, narration and rendering. Added deterministic plan checks, repair and post-render validation of streams, resolution and duration.

SELECTED ACADEMIC PROJECTS

Smart Summarizer | Language Model Fine Tuning
- Fine-tuned language models with LoRA, PyTorch and Hugging Face for academic-paper summarization; evaluated summaries using ROUGE, BLEU, BERTScore and LLM-as-Judge.

Tuberculosis Screening from Cough Audio | Final Year Project
- Integrated XGBoost, a spectrogram model and Wav2Vec2 through weighted voting, with audio preprocessing and parallel inference. Reported 98.33% accuracy in the project evaluation; placed third among 250 teams at FAST Job Fair 2025.

UNIVERSITY TEACHING EXPERIENCE

FAST-NUCES Lahore — Lahore, Pakistan
AI Lab Assistant — Jan 2025 - May 2025

Teaching Assistant in Parallel and Distributed Computing — Aug 2024 - Dec 2024
- For Parallel and Distributed Computing, supported two undergraduate sections through scheduled student office hours and weekly coordination with the course instructor.

RESEARCH PUBLICATION

Co-author, MultiObjective Optimization in Wireless Mesh Networks - A Survey. ACM Computing Surveys, 2026. DOI: 10.1145/3800685.

TECHNICAL SKILLS

AI and multimodal: PydanticAI, LangGraph, LoRA, PyTorch, Hugging Face, OpenAI Realtime, LiveKit, Whisper, YOLOv8, TensorFlow Lite.

Retrieval and data: Embeddings, BM25, ChromaDB, pgvector, Pinecone, PostgreSQL, Firestore, Redis.

Engineering and delivery: Python, TypeScript, FastAPI, Django, React, Flutter, Celery, Docker, GitHub Actions, pytest, AWS Textract.

Spoken languages: English - fluent (IELTS 7.5); Urdu - native; Swedish - beginner.